

1 Introduction

We study *probability theory*, which is motivated by the analysis of *uncertainties* arising in physical and social sciences and daily life, as a subfield of mathematics. By *mathematics*, we mean logical systems built on axioms, definitions and logical deductions. Mathematics provides a systematic approach to express ideas, as well as their logical dependence, efficiently and unambiguously. The definitions and theorems in probability theory are primarily motivated by chance events and processes such as coin tossing, motion of “particles” (e.g., stock prices), plausible reasoning, etc. The resulting theory can then be used to model actual chance events and perform inference (see Figure 1.1), but the validity of theorems in probability theory does *not* depend on the interpretations of probability. Nowadays, probability theory is widely applied in various fields including statistics and machine learning.

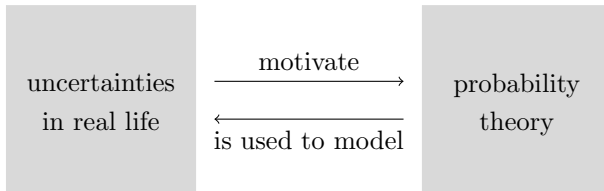


Figure 1.1: Interactions between mathematical probability theory and its applications.

1.1 Interpretations of probability

Intuitively, probability is a *quantification of uncertainty* which may arise in many ways.¹ There are two major interpretations of probability: *frequentist* and *Bayesian*. Remarkably, both interpretations lead to the same mathematical properties that probabilities should satisfy.

1.1.1 Frequentist probability

Suppose we flip a coin 20 times and observe 12 heads. The *relative frequency* of heads is $\frac{12}{20} = 0.6$. According to the relative frequency interpretation of probability, to say that the probability of heads is 0.5 means the following: if we flip the coin many, many times “under the same condition”, the relative frequency of heads approaches 0.5. In general, the frequentist interpretation of the probability of an event A is

$$\Pr(A) = \lim_{n \rightarrow \infty} \frac{\# \text{ of times } A \text{ occurs in } n \text{ trials}}{n}. \quad (1.1)$$

As the limiting value of a fraction, $\Pr(A)$ lies between 0 and 1. Note that here we use the symbol $\Pr$ to distinguish this concept from the *probability measure* $\mathbb{P}$ (Definition 2.7) in the mathematical framework.

¹See Section A.1 for more discussion about the concept of uncertainty.

Suppose A and B are *mutually exclusive* events, i.e., if A occurs B cannot occur and vice versa. Then

$$(\# \text{ of times } A \text{ or } B \text{ occurs}) = (\# \text{ of times } A \text{ occurs}) + (\# \text{ of times } B \text{ occurs}).$$

From (1.1), we expect that for mutually exclusive events we have

$$\Pr(A \text{ or } B) = \Pr(A) + \Pr(B).$$

This *additive property* will be captured in the mathematical theory. The frequentist approach is appropriate and useful in many applications. An advantage is that this approach is more objective when the “experiment” is well-defined and replicable; it also suggests that one can estimate probabilities by repeated sampling. On the other hand, it requires that the experiment can be repeated indefinitely under the same condition, at least in principle.

1.1.2 Bayesian probability

According to the Bayesian interpretation, probability is *subjective* and measures the degree of belief of a person or computer with some state of knowledge. In particular, the chance event does *not* have to be repeatable. With 0 representing “impossibility” and 1 representing “certainty”, the probability of an uncertain event lies somewhere between 0 and 1. When we obtain new information, the probability is updated via *Bayes’s rule*.

Closely related to the Bayesian approach is the viewpoint that probability theory is an *extension of logic*. Ordinary logic is concerned with *deductions* such as this one:

- Suppose A is true and A implies B . Then B is true.

Deductive logic, which is binary (true/false), cannot deal with uncertain events: if we don’t know whether A is true, we don’t know whether B is true even if it is given that A implies B . Bayesian probability, on the other hand, allows *inductions* such as the following one:

- Suppose A implies B and we observe that B is true. Then, A becomes more plausible.

For example, Einstein’s theory of general relativity implies the existence of gravitational waves which were observed in 2015. This gave physicists more confidence in the validity of general relativity. The mathematical rules of probability (sum and product rules) then correspond to *consistency* of the extended logic, e.g., the probabilities should not depend on the way we carry out the induction and are consistent with “common sense”. In fact, it can be shown that if plausibilities are quantified by numbers and satisfy certain reasonable properties, then they must satisfy the sum and product rules of probability.²

As an extended logic, Bayesian probability can potentially be applied to a wide range of phenomenon including those where the frequentist approach is inappropriate. However, it is generally not easy to translate a given state of knowledge into probability values.

²This is the content of *Cox’s theorem* (https://en.wikipedia.org/wiki/Cox_theorem).

1.2 Overview of probability theory

We summarize briefly the structure of probability theory. A chance event, also referred to as an *experiment*, is modelled mathematically by choosing (i) a set Ω of possible outcomes called the *sample space*, and (ii) a *probability measure* $\mathbb{P}$ on Ω which specifies the probability $\mathbb{P}(A)$ with which each event A occurs. For example, in Table 1.1, we imagine tossing three coins and each of the 8 possible outcomes is assumed to occur with probability $\frac{1}{8}$.

Prob	Outcome	Value of X (# of +)
1/8	+++	3
1/8	++-	2
1/8	+ - +	2
1/8	+ - -	1
1/8	- + +	2
1/8	- + -	1
1/8	- - +	1
1/8	- - -	0
		1/8 3/8 3/8 1/8

Table 1.1: A probabilistic model for tossing 3 coins.

Any probability measure $\mathbb{P}$ must satisfy the following properties:

- *Normalization:* $\mathbb{P}(\Omega) = 1$, i.e., the “total mass” is 1.
- *Sum rule:* If $A_1, A_2, \dots$ are mutually exclusive then $\mathbb{P}(\bigcup_i A_i) = \sum_i \mathbb{P}(A_i)$.
- *Product rule:* For any events A, B we have, by definition, $\mathbb{P}(A \text{ and } B) = \mathbb{P}(A)\mathbb{P}(B|A) = \mathbb{P}(B)\mathbb{P}(A|B)$, where $\mathbb{P}(A|B)$ is the *conditional probability* of A given B . This implies, e.g., $\mathbb{P}(ABC) = \mathbb{P}(A)\mathbb{P}(B|A)\mathbb{P}(C|AB)$ where AB means “ A and B ” and so on.

Given $(\Omega, \mathbb{P})$, to be referred to as a *probability space*, we can study various *random variables* which depend on the outcome. For example, let X be the number of +. Then $\mathbb{P}(X = 2) = \mathbb{P}\{++-, +-+, -++\} = \frac{3}{8}$ (see (2.1)). The probabilities $\mathbb{P}(X = k)$, for all k , determine the *distribution* of X .

Now we may summarize some major topics of basic probability theory:

- Methods of *computing probabilities* (e.g., law of total probability, Bayes’s rule, permutations and combinations).
- *Special random variables and distributions* which are important in theory and application (e.g., binomial, Poisson and normal distributions).
- Concepts for *describing random variables and distributions* (e.g., probability mass/density functions, expected value, generating functions).
- *Relations* among a collection of random variables (e.g., joint and conditional distributions, independence, covariance and correlation).
- *Construction of new random variables/processes* from given ones (e.g., partial sums, sample means, ratios and nonlinear transformations).
- *Limiting behaviours*, especially when a large number of random variables interact with each other under suitable structural conditions.